

GRAND TRAVERSE LIGHT, MI, USA

WMO: 997731

Lat: 45.2106N

Lon: 85.5503W

Elev: 604

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.7	7.8	N/A	N/A	N/A	N/A	N/A	N/A	42.2	29.1	38.3	23.0	14.2	280	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	82.0	N/A	79.3	N/A	77.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	10.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.2	31.6	27.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	23.8	23.0	30.2	40.4	50.9	60.9	68.2	69.0	62.7	51.3	40.0	31.6
	DBStd	17.69	8.30	8.82	9.80	7.32	7.93	6.85	5.00	4.42	6.14	7.15	7.69	6.80
	HDD50	3522	811	757	619	306	81	2	0	0	0	66	310	570
	HDD65	7246	1276	1177	1079	739	446	160	27	13	115	430	749	1035
	CDD50	2114	0	0	5	17	109	330	563	590	381	108	11	0
	CDD65	361	0	0	0	0	9	37	125	137	47	6	0	0
	CDH74	1283	0	0	1	1	72	158	455	428	150	18	0	0
	CDH80	181	0	0	0	0	15	34	54	57	19	2	0	0
Wind	WSAvg	11.6	15.5	13.8	11.1	10.8	8.7	7.7	9.3	10.1	11.0	13.2	13.5	14.3
Precipitation	PrecAvg	31.3	2.1	1.5	1.8	2.6	3.0	2.8	2.9	3.0	3.5	3.1	2.7	2.2
	PrecMax	37.1	4.1	2.9	4.4	4.6	5.5	7.4	6.7	5.1	6.6	7.0	5.2	5.2
	PrecMin	24.3	0.7	0.6	0.5	0.6	1.3	0.6	1.1	0.9	0.7	1.1	1.2	0.5
	PrecStd	3.1	0.9	0.6	1.0	1.0	1.1	1.5	1.4	1.2	1.5	1.6	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	44.4	66.7	72.7	82.6	85.4	85.7	85.3	82.6	76.4	64.0	54.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	42.2	38.6	54.8	64.3	75.0	78.7	81.0	81.3	78.2	70.2	57.0	45.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	38.5	36.0	47.4	57.5	69.0	74.5	78.3	78.4	74.9	65.4	53.1	43.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	35.7	34.2	42.2	51.4	64.0	71.1	76.5	76.2	72.0	61.2	50.1	41.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.5	12.6	12.7	16.0	14.4	13.1	12.4	12.9	10.6	9.0	7.5
		MCDBR	11.8	11.6	19.4	23.2	24.6	19.2	15.4	14.8	16.4	17.4	14.0	10.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.288	0.293	0.319	0.354	0.385	0.395	0.391	0.385	0.372	0.342	0.307	0.282
	taud		2.433	2.377	2.391	2.357	2.349	2.358	2.392	2.408	2.427	2.485	2.520	2.489
	Ebn at Noon		264	284	290	287	280	277	277	274	269	262	255	255
	Edh at Noon		23	29	32	37	38	38	36	35	31	26	21	19
All-Sky Solar Radiation	RadAvg		373	668	1125	1459	1782	1938	1953	1660	1277	747	403	275
	RadStd		39	69	119	147	153	139	78	105	110	88	57	29

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page